DATA ANALYTICS PROJECT

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments & product preferences from **3,900 transactions**

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CHAPTER 1

Dataset at a Glance

3,900	18	50	25
Transactions	**Data Columns**	**Locations**	**Product Types**

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discounts, Promos, Frequency, Ratings, Shipping

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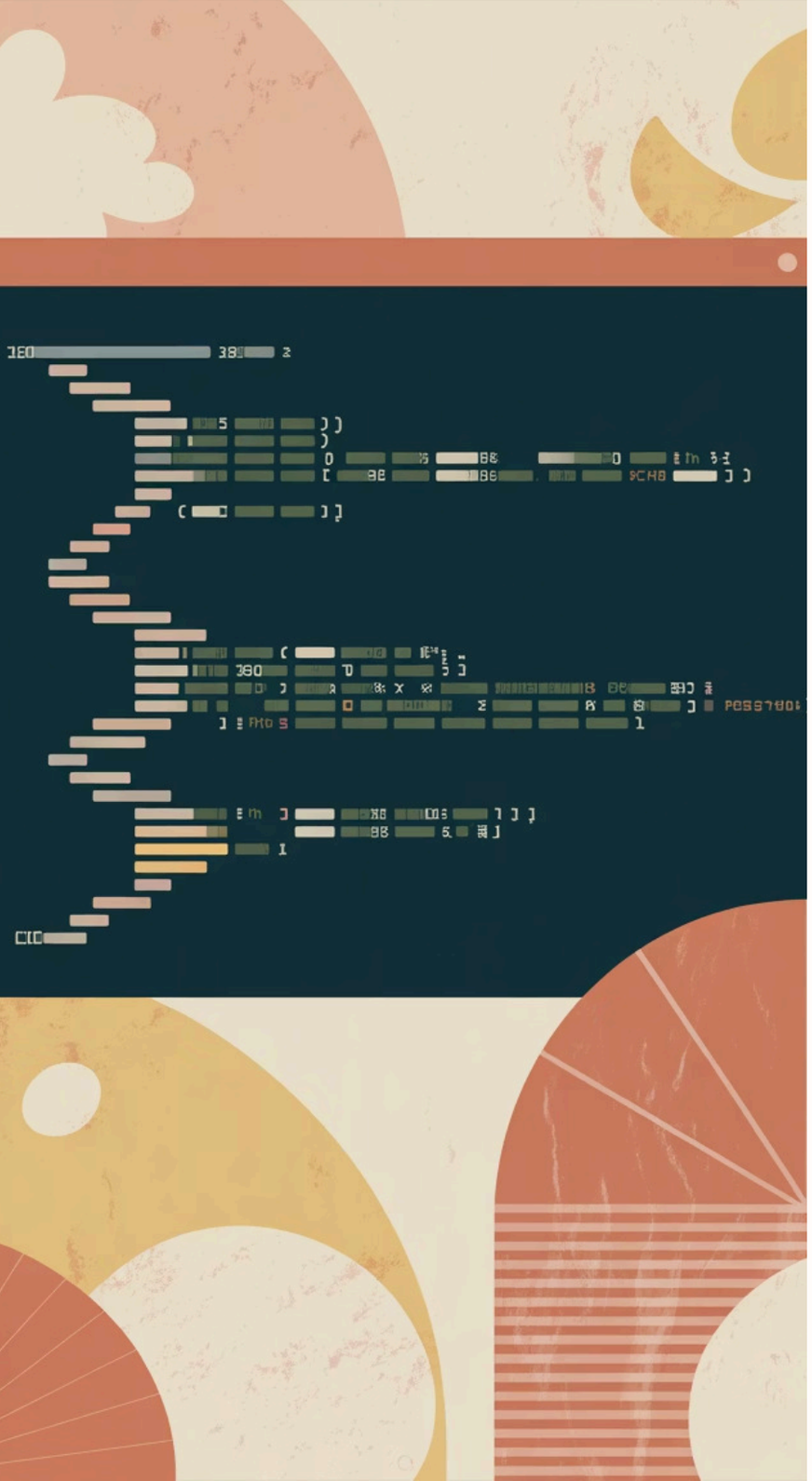

CHAPTER 2

Exploratory Data Analysis in Python

01

Data Loading & Exploration

Imported with `pandas`, checked structure via `df.info()`

02

Missing Data Handling

Imputed 37 missing Review Ratings using median per category

03

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

04

Cleanup & Export

Dropped redundant `promo_code_used`, loaded to PostgreSQL

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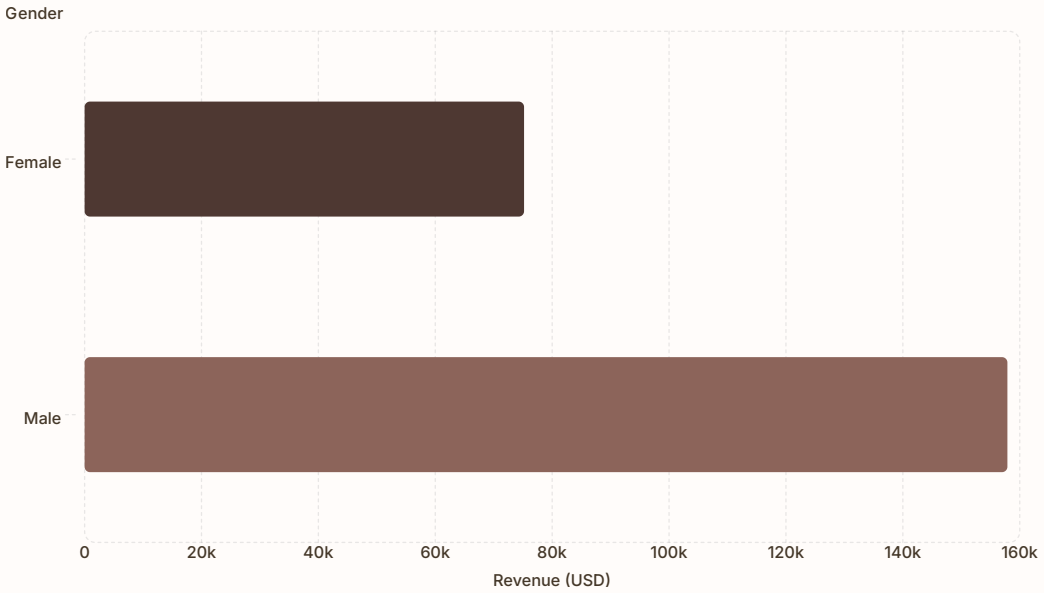
Revenue & Spending Insights



Revenue by Gender

Male customers generated **2x more revenue** ($157,890 vs $75,191)

High-Spending Discount Users

839 customers used discounts yet spent above the average purchase amount of $59.76

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Product Ratings & Shipping

Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type Comparison

Standard

Avg $58.46

Express

Avg $60.48

Express users spend slightly more per transaction

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Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Subscribed	1,053	$59.49	$62,645
Non-Sub	2,847	$59.87	$170,436

Avg spend is nearly identical — but only **27%** of customers subscribe, leaving a huge conversion opportunity.

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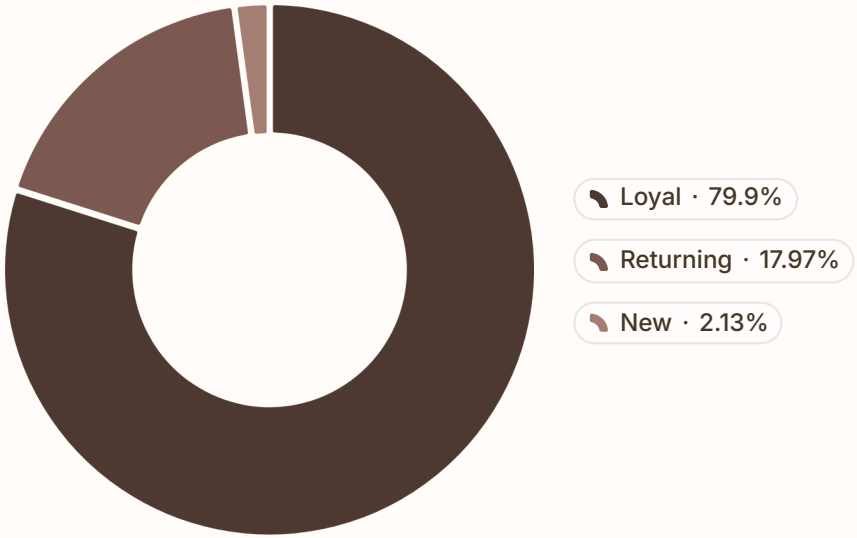
Customer Segments & Discount Dependency



Customer Segmentation

80% are Loyal, 18% Returning, 2% New

Top Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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Power BI Dashboard

3.9K

Customers

$59.76

Avg Purchase

3.75

Avg Rating

73%

Non-Subscribers

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CHAPTER 5

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers

Loyalty Programs

Reward repeat buyers to grow the "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control on high-discount products

Product Positioning

Highlight top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

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